

HF-IQR V3 — Pre-Registration

Title: HF-IQR V3: A Pre-Registered Confirmatory Study of GPT-4o Critique Behavior Under Multi-Round LLM Deliberation

Researcher: Billy Davis — Independent Researcher, Hudson Forge IRMB-C, Lenoir, NC

Status: Filed pre-registration. This document is registered on the Open Science Framework (OSF) prior to data collection. Per the zero-amendment commitment, the design described here is fixed; any unforeseen issue not covered by a pre-registered branch will be reported transparently as a disclosed deviation.

Filing venue: Open Science Framework (OSF), open-ended registration.

1. Study Purpose and Background

1.1 Background

HF-IQR V2 (pre-registered, five frontier models, 200 questions, 5,137 responses) produced a behavioral observation about GPT-4o: across Round 2 anonymous peer critique, GPT-4o produced **validation responses rather than critiques**, with a mean Critique Validity Score (CVS) of 0.433 against 0.573–0.644 for the other four models (CVS ANOVA $F(4,991)=47.81$). V2 characterized this as a behavioral finding and reported it as the strongest contribution of that study, alongside the Position Stability / Correctness Rate dissociation.

V2 also documented limitations: CVS inter-rater reliability fell below threshold ($r=0.524$ – 0.569), the ESVR metric was abandoned after a validation failure (local models could not reliably evaluate frontier reasoning), and three pre-registration deviations were filed as post-hoc amendments.

1.2 Purpose of V3

V3 has a single, narrow confirmatory purpose. It is important to state that purpose precisely, because V3 is **not** a direct replication of V2's natural-behavior finding.

V2 observed GPT-4o's critique behavior under *ordinary* anonymous peer critique — i.e., with no constraint on whether the model validated or critiqued. V3's Round 2 deliberately changes that: it employs a forced counterargument protocol that structurally blocks validation language (Section 3.3). V3 therefore cannot, and does not claim to, test whether GPT-4o *naturally* validates instead of critiques — that behavior is suppressed by design in Condition B.

What V3 confirmatorily tests is narrower and is stated as such: **whether GPT-4o still exhibits lower critique validity than the other four frontier models when validation language is forcibly blocked.** In other words, V3 tests whether GPT-4o's critique-quality deficit *persists under a forced-counterargument protocol*, rather than whether the deficit appears under unconstrained conditions. This is a legitimate confirmatory question and arguably a stronger one — it asks whether the deficit is robust to an intervention designed to remove its most obvious mechanism — but it is a *different* question from V2's, and the prereg treats it as such. V3 is not a comprehensive benchmark and does not aim to expand category coverage.

1.3 Relationship to prior published work

The multi-agent deliberation literature (multi-agent debate; CriticBench; controlled debate studies) overwhelmingly studies deliberation as a route to improved accuracy. V3 differs in three combined respects: it is a pre-registered confirmatory study; it isolates a specific per-model behavioral signature rather than aggregate accuracy; and it uses an explicit causal control condition. A full related-work section will accompany the final paper.

2. Hypotheses

2.1 Primary confirmatory hypothesis (H1)

Under the V3 deliberation protocol (Condition B) — in which Round 2 employs a forced counterargument protocol that structurally blocks validation language — GPT-4o will exhibit a lower mean composite Round 2 CVS than the other four frontier models. H1 tests whether GPT-4o's critique-quality deficit *persists under this intervention*; it is not a test of GPT-4o's unconstrained natural critique behavior (see Section 1.2).

H1 is confirmed only if BOTH of the following pre-registered criteria are met: - (a) the contrast is statistically significant at the pre-registered alpha level (Section 6); AND - (b) the effect size meets or exceeds the pre-registered minimum effect-size floor of Cohen's $d \geq 0.55$ (Section 6).

A result that is statistically significant but below the effect-size floor is recorded as NOT confirming H1. The floor is set at $d \geq 0.55$, the lower bound of the 95% confidence interval of V2's own GPT-4o-vs-others CVS effect (V2: $d = 0.64$, 95% CI [0.55, 0.74]; see Section 6.2). Tying the floor to V2's CI lower bound is an established replication standard: it asks whether V3's effect is consistent with the smallest effect V2's data credibly supports, rather than imposing an arbitrary threshold.

2.2 Secondary analyses

The following are pre-specified as secondary / exploratory. They are labeled as such in the final paper and CANNOT be reported as confirmatory findings (see Section 7, Exploratory Analysis Registry).

- The causal contrast: Condition A vs Condition B effect on CVS. This tests a question V2 did not pose (whether deliberation *causes* the behavior) and is therefore exploratory-confirmatory, not a replication.
- The three CVS sub-dimensions analyzed independently.
- Round 5 revision-trigger category distributions.
- Per-category reasoning effects across the 12 categories.
- Round 1 confidence-calibration analysis.
- Item-level Position Stability / Correctness Rate relationship.
- Exploratory-arm added models (Section 3.4).

3. Design

3.1 Subject models — confirmatory pool (frozen)

The confirmatory pool is frozen to the five V2 models for exact historical alignment:

- claude-sonnet-4-5 (Anthropic)
- gpt-4o (OpenAI)
- gemini-2.5-pro (Google)

- deepseek-chat (DeepSeek)
- grok-3 (xAI)

These are provider **alias** identifiers (not pinned dated snapshots), retained identical to V2 for continuity. Alias-based APIs may resolve to updated underlying model versions over time, so exact version-identity with V2 cannot be guaranteed; this is a documented limitation of replicating against alias-based commercial APIs. V2 recorded only the aliases, so the snapshots V2 resolved to are not recoverable. To make V3 itself reproducible, V3 logs the resolved model version returned in each API response. Provider endpoints are recorded in a supplemental table at filing.

3.2 Conditions — causal design

- **Condition A (no deliberation):** Round 1 only.
- **Condition B (full protocol):** all five rounds.

Both conditions are run over the IDENTICAL question set. Both conditions apply the IDENTICAL Round 1 JSON output schema. The only difference between A and B is the presence of Rounds 2-5. This isolates the deliberation protocol as the manipulated variable.

Condition A and Condition B are executed as **separate, stateless runs** with no shared session, context, or memory between them. Because the runs share no state, run order does not exist as a variable and no counterbalancing is required. The pre-flight gate (Section 8) verifies that the two runs are genuinely independent with no leaked context.

3.3 Round protocol

All rounds enforce a strict JSON output schema. Free-text drift is not permitted.

Round 1 — Independent response. Each model answers each question independently. Output schema:

```
{
  "primary_claim_validation": boolean,
  "confidence_calibration_score": float,    // 0.0-1.0
  "rationale_prose": string
}
```

The prompt explicitly anchors the confidence scale: 0.0 = completely uncertain, 1.0 = completely certain, to reduce model-specific scaling.

Round 2 — Anti-validation critique (the validation-prevention intervention). Round 2 replaces generic peer critique with **forced counterargument generation**. The system prompt instructs each model to produce the three strongest, structurally independent arguments against the position under review. Affirmative or validating language is explicitly prohibited at the prompt level. If a model judges that no strong counterarguments exist, it must explicitly state this and then still generate the three most plausible counterarguments a skeptic would raise — silence or refusal is not a permitted output, as a refusal would itself constitute a validation response. (Consistent with the rubric convention: the protocol measures whether targeted counterarguments are generated, not whether they are ultimately correct.)

This intervention bundles two changes (forced counterargument generation + affirmative-language prohibition) and is pre-registered as a SINGLE "anti-validation intervention." V3 tests the bundle, not its component parts. Forced counterargument generation will sometimes produce weak objections against correct claims; this is expected, is symmetric across all five models, and is modeled as a uniform inflation of the false-positive critique rate, causing no differential bias to the between-model contrast.

Round 3 — Defend or revise. Each model receives the critique of its own Round 1 response and states whether it defends or revises, with justification.

Round 4 — Mirror self-assessment. Each model compares its response to ground truth and an anonymous peer response and assesses its own accuracy and calibration.

Round 5 — Mechanistic trace (split output). Each model explains why it defended or revised. Output is split into two logged JSON fields:

```
{
  "self_assessment": float,           // 0.0-1.0
  "revision_trigger_category": string // "logical" | "social" |
  "uncertainty"
}
```

The revision-trigger taxonomy is the three-category V2 taxonomy (logical / social / uncertainty), kept identical to V2 to preserve confirmatory comparability

with V2 Finding 3. A proposed fourth category ("structural") was considered and deferred to a future study, as adding a category would break the V2 comparison baseline.

3.4 Exploratory model arm

Models outside the frozen pool (e.g., Mistral Large 3, `mistral-large-2512`, and others) may be run through the full protocol as a SEPARATE exploratory arm. Their data is reported separately and is NEVER pooled into the confirmatory analysis, to prevent confounding model identity with model class. Exact IDs pinned at filing. If a newer variant of an exploratory model is released before filing, the model-ID string is updated once by explicit researcher decision (not by automated substitution).

3.5 Sampling parameters

All five models are set to their **provider-defined minimum randomness setting** (referred to as "temperature 0.0" for brevity) across all five rounds, to maximize reproducibility. Where an API does not permit exactly 0.0, its documented minimum is used and recorded per model. A supplemental table of exact sampling parameters per model is included at filing. Note that "temperature 0.0" is a nominal label: providers implement minimum-randomness settings differently and some retain internal nucleus sampling, so exact determinism is not assumed across providers. The actual sampling parameters returned by each provider are logged from raw API responses and recorded in the supplemental table, so compliance is verifiable rather than asserted.

3.6 Dataset

The question set spans the 12 reasoning categories established in V2. Ground truth is provided for all questions; verifiable binary ground truth is required for the categories used in correctness scoring. Expert-verified subsets are used for complex categories. The question count is fixed at **N = 80 questions**, distributed across the 12 reasoning categories. This value is derived from the power analysis (Section 6.3); it was not pre-committed but follows from the analysis.

4. Primary Outcome Measure

The primary outcome is the **composite Round 2 Critique Validity Score (CVS)**, constructed comparably to V2 for replication validity.

For confirmatory reporting, the primary outcome is the composite CVS. The three independent sub-dimensions (logical_coherence, critique_actionability, stylistic_bias — Section 5) are secondary and serve to locate where the behavior appears; they do not constitute the confirmatory test.

5. Scoring Methodology

5.1 CVS dimensions

CVS is scored across three independent dimensions, each scored separately on a 0–1 float scale, with no pooling into a single number prior to analysis (to avoid halo-effect correlation): - logical_coherence - critique_actionability - stylistic_bias

The composite CVS (primary outcome) is derived from these three dimensions using the same weighting as V2's CVS construction, for replication comparability. The exact V2 dimension weights are recorded in the supplemental table; V3 does not revise them, as changing the weighting would break comparability with the V2 effect against which V3 is assessed.

The full CVS scoring rubric is attached as a supplement to this pre-registration (file: HF-IQR-V3_CVS_Rubric_Supplement) and is fixed before data collection. The supplement adapts V2's five CVS criteria into the three reporting dimensions (logical_coherence, critique_actionability, stylistic_bias), uses binary item scoring, preserves V2's weights, and provides per-item definitions and anchor examples for rater training. One verification item remains: confirming the rubric matches V2's CVS criteria as actually applied (to be done in the final cold-pass review).

If, under a multi-rater tier (Section 5.3), the inter-rater agreement statistic drops by more than 0.05 between the pilot and the main validation, scoring halts and raters are retrained (in addition to the reliability-failure branch of Section 5.4).

5.2 Scoring reliability gate

CVS scores are admissible for confirmatory analysis only if they clear the inter-rater reliability gate in Section 5.4. Scores from any dimension that fails the gate are demoted to exploratory.

5.3 Scoring architecture — model-primary with human validation

V3 uses a **model-primary scoring architecture with human validation**. This is a deliberate design choice for an independent research program without the resources to recruit and calibrate a full human rater panel. The architecture and its safeguard are pre-registered here.

Primary scorer. A cloud-hosted model (e.g., `mistral-large-2512` via API), running in isolation, scores the full critique dataset on the three CVS dimensions. Model scoring is the primary measurement process for V3 — not a fallback. This is disclosed plainly: the primary CVS measure is model-generated, and its trustworthiness is established through the human validation procedure below rather than assumed.

Human validation subset. A bounded subset of critiques (the "validation subset") is scored independently by human rater(s). The validation subset is NOT scored by models — it is the human anchor against which the model-scorer is checked. The model-scorer's scores on the same subset are compared to the human scores via the two-axis decision-surface test of Section 6.6 (TOST on Delta-mu, Levene's test on Delta-sigma). This comparison is what licenses or withholds confirmatory status for H1.

The validation subset is **50 Round 2 critiques**, sampled **stratified** across all 12 reasoning categories and all five confirmatory-pool models, so that validation is not concentrated on easy items or any single model. Fifty is large enough for a stable agreement statistic and a meaningful TOST / Levene comparison, and small enough to be completed by a one- or two-rater tier.

Rater tier — pre-registered to match realistic recruitment. Human rater recruitment for unpaid annotation is difficult; V3 pre-registers a tiered validation that scales honestly to the number of raters obtained:

- **Three or more raters:** inter-rater reliability on the validation subset is reported with Krippendorff's alpha, and the alpha-failure branch of Section 5.4 applies.

- **Two raters:** reliability is reported with Cohen's kappa (and percent agreement). This is pre-registered as a *limited* reliability indication, weaker than a full multi-rater analysis, and is disclosed as such.
- **One rater (researcher only):** the researcher scores the validation subset blind. This is pre-registered honestly as a single-expert reference check, explicitly labeled a limitation; no inter-rater statistic is computed.

The realized tier is recorded at the time of validation. The study proceeds under whichever tier is reached; there is no recruitment deadline gating the whole study, because model scoring is primary by design.

Raters are human. The validation subset must be scored by humans. LLMs (including any council models) are explicitly excluded from the rater role. Models validating a model would be circular and would repeat the V2 ESVR failure, in which models could not reliably evaluate frontier reasoning (Section A8). The council's role is design review only, never validation scoring.

Validation outcome governs H1 status (resolves former P1a/b/c). - The model-scorer is "validated" if, on the validation subset, the two-axis test (Section 6.6) does not show Directional Displacement (Drift) relative to the human anchor. The model-scorer may widen uncertainty (Endpoint B) and still pass; systematic Drift (Endpoint A) constitutes validation failure. - If the model-scorer passes validation, H1 retains confirmatory status: the confirmatory object is the *validated measurement process*, not the species of scorer. - If the model-scorer fails validation (Drift relative to the human anchor), H1 downgrades to exploratory — the interpretation fails, not merely the scorer. - Under the single-rater tier, the validation is correspondingly weaker; the prereg discloses that a solo-validated result carries a weaker confirmatory claim than a multi-rater-validated one, and the final paper states which tier was realized.

5.4 Annotation reliability — reliability-failure branch

This branch governs the **3-or-more-rater tier** (Section 5.3), in which inter-rater reliability is measured by Krippendorff's alpha. Per scoring dimension: - **alpha $\geq$ 0.75:** the dimension proceeds to confirmatory analysis. - **0.67 $\leq$ alpha < 0.75:** halt; revise the scoring rubric exactly once; retrain raters; re-pilot on a subset of 15 items. If the second pass reaches alpha $\geq$ 0.75, proceed. If not, the dimension is demoted to exploratory. - **alpha < 0.67:** immediate demotion of that dimension to exploratory; it is excluded from the confirmatory analysis.

Because the re-pilot subset is small, its alpha estimate carries wider uncertainty; a borderline re-pilot result is interpreted conservatively (a result at the threshold is treated as not clearing it).

Under the two-rater tier, the same thresholds are applied to Cohen's kappa in place of alpha, with the explicit disclosure (per Section 5.3) that two-rater agreement is a limited reliability indication. **Under the single-rater tier**, no inter-rater statistic exists; reliability cannot be gated, and the confirmatory claim is correspondingly weakened and disclosed as such in the final paper. In all tiers, the model-scorer validation of Section 6.6 still applies — it does not depend on rater count.

5.5 Revision-category validity check (O8)

During the pilot, revision traces are classified twice: once by the model (self-report, per protocol) and once by an independent blind human classification (the researcher). A confusion matrix compares the two. - Model-human agreement ≥ 0.75 : model self-classification is trusted; R5 proceeds as designed. - Below 0.75: human classification becomes the primary source for the revision-trigger analysis; model self-report is demoted to secondary/exploratory. - If the confusion matrix shows two categories systematically confused, that is documented as a limitation and those categories may be reported merged. The 0.75 threshold is set to match the reliability standard used elsewhere in the study (the alpha gate of Section 5.4 at the multi-rater tier); here it is applied as a model-vs-human agreement rate on the revision-category classification.

6. Statistical Framework

All tests are pre-specified here before data collection.

6.1 Primary confirmatory test

The H1 contrast (GPT-4o composite Round 2 CVS vs the other four models, Condition B) is the single primary confirmatory test. As a single pre-specified test it is reported WITHOUT multiple-comparison correction.

- Significance threshold (alpha): 0.05.
- Effect-size floor for confirmation: Cohen's $d \geq 0.55$. Rationale in Section 6.2.

6.2 Effect size for power analysis

The expected effect size for the power analysis is derived from V2's own CVS data — specifically, the effect size for the GPT-4o-vs-others contrast, recovered from V2's published CVS means (GPT-4o 0.433 vs the 0.573-0.644 cluster) and their variance. It is contrast-matched to the H1 comparison rather than taken from the omnibus V2 F-statistic, and it is NOT discounted or halved. The contrast-matched effect size, computed from V2's CVS data, is **Cohen's $d \sim 0.64$** (GPT-4o mean CVS 0.433 vs other-four mean 0.604; pooled SD ~ 0.268). Per-model standard deviations were derived from V2's 95% bootstrap confidence intervals (V2 notebook cell 148), so the pooled SD is approximate; this is noted as a minor limitation. The power analysis (Section 6.3) uses $d = 0.64$ as its input effect size.

6.3 Power analysis

The sample size targets the H1 confirmatory contrast (GPT-4o composite Round 2 CVS vs the other four models, Condition B) — not descriptive statistics — at $\alpha = 0.05$ and target power = 0.80.

The input effect size is taken conservatively at **$d = 0.55$** , the lower bound of the 95% CI of V2's GPT-4o-vs-others CVS effect (V2: $d = 0.64$, 95% CI [0.55, 0.74]; Section 6.2). Sizing to the CI lower bound rather than the point estimate protects against underpowering if the true effect is at the weaker end of what V2's data supports.

A two-group power calculation at $d = 0.55$, $\alpha = 0.05$, power = 0.80 yields approximately 53 observations per side. Because V3's confirmatory test is a mixed-effects model with questions crossed with models, an item-level random effect, and an unbalanced 1-vs-4 (GPT-4o vs others) contrast, a design-effect adjustment of 1.5x is applied to account for the clustered structure. This yields **$N = 80$ questions**.

The design-effect multiplier (1.5x) is itself an estimate; the true intra-class correlation is not known until data exists. The 1.5x value is pre-registered as a stated, conservative assumption rather than a precise quantity. The achieved power is reported in the final paper against the realized data structure.

6.4 Secondary / exploratory analyses

All secondary and exploratory analyses (Section 2.2, Section 7) use the Benjamini-Hochberg False Discovery Rate (FDR) procedure for multiple-comparison control. FDR is chosen over Bonferroni to avoid excessive Type II error inflation across the multi-dimensional behavioral measures.

Revision-trigger analysis uses a categorical mixed-effects regression model, replacing V2's keyword pattern matching. The Position Stability / Correctness Rate relationship is computed at the individual item level (not aggregated to $n=5$ model means, which was a documented V2 limitation). It is pre-registered as a mixed-effects logistic regression: `Correctness ~ PositionStability + Model + Category + (1 | Item)`. The PositionStability coefficient is reported as suggestive/exploratory only.

6.5 Missing data

Per the malformed-output rule (Section 8), a model response that fails schema validation after one retry is recorded as missing data — not imputed, not substituted. The mixed-effects models handle unbalanced cells natively.

6.6 Model-scorer validation — the two-axis decision surface

The model-scorer (primary, Section 5.3) and the human validation subset are compared on the same 50 critiques. The model-scorer is characterized on two orthogonal axes — change in central tendency (Delta-mu) and change in dispersion (Delta-sigma) — rather than by a single overloaded test. Direction and uncertainty are treated as independent coordinates, not competing explanations.

Endpoint A — Directional Displacement (Drift). Prediction: location changes while dispersion stays approximately stable. The mean difference between model-scored and human-scored CVS is tested in the mixed-effects model; the effect estimate and confidence interval are reported. Whether Delta-mu counts as "approximately 0" is decided by two one-sided tests (TOST) against the pre-registered Delta-mu equivalence bound: a result inside the bound is treated as no meaningful directional movement; a result outside it is directional displacement. Decision: Delta-mu outside the equivalence bound and Delta-sigma within its bound -> Drift.

Endpoint B — Symmetric Uncertainty Expansion. Prediction: dispersion increases while central tendency stays approximately stable. The equality of

variance between model-scored and human-scored CVS is tested with Levene's test; the magnitude of any dispersion change is reported as a relative change against the pre-registered Delta-sigma bound. Decision: Delta-mu within its equivalence bound (by TOST) and Delta-sigma outside its bound (by Levene's test) -> Uncertainty Expansion.

The statistical procedures (TOST for Delta-mu, Levene's test for Delta-sigma) are pre-registered here. The equivalence bound values they test against are set separately from pilot data (see below) - the tests are fixed in advance; only the bound magnitudes await the pilot.

The four-cell decision surface. All outcomes are pre-assigned a cell, using the TOST result for Delta-mu (inside vs outside its equivalence bound) and the Levene's-test result for Delta-sigma (within vs outside its bound): - Delta-mu outside bound, Delta-sigma within bound -> Drift - Delta-mu within bound, Delta-sigma outside bound -> Uncertainty Expansion - Delta-mu outside bound, Delta-sigma outside bound -> Mixed (movement + destabilization) - Delta-mu within bound, Delta-sigma within bound -> Null

Every result lands in a named cell; no outcome is uninterpretable.

Equivalence bounds as governance primitives. The bounds defining "approximately 0" for both Delta-mu and Delta-sigma are pre-declared here, before any data is seen, as principled statements of what magnitude of change is scientifically meaningful — NOT as convenience thresholds. The cells inherit their meaning from these bounds, so the bounds must be justified, not merely chosen. [TODO – predeclare the Delta-mu equivalence bound and the Delta-sigma equivalence bound, each with a written rationale, before filing. Set with Billy as a focused decision; the Delta-mu bound is a candidate to anchor to the $d \geq 0.55$ effect-size floor.]

Validation anchor. The model-scorer is "validated" if, on the validation subset, it does not produce Directional Displacement relative to the human comparison anchor — i.e., the model-scorer may widen uncertainty (Endpoint B) and still pass, but systematic Drift (Endpoint A) relative to human scoring constitutes validation failure. Per Section 5.3, validation-anchor failure — not the use of a machine scorer — governs whether H1 downgrades to exploratory.

7. Exploratory Analysis Registry

The following exploratory analyses are registered in advance. Any analysis NOT on this list at filing time may be reported only as a hypothesis for a future study — never as a V3 finding.

1. Causal A vs B contrast (deliberation effect on CVS).
2. The three CVS sub-dimensions analyzed independently.
3. Round 5 revision-trigger category distributions.
4. Per-category reasoning effects across the 12 categories.
5. Round 1 confidence-calibration analysis (e.g., Brier score, expected calibration error).
6. Item-level Position Stability / Correctness Rate relationship.
7. Exploratory-arm added-model results (Section 3.4).

This registry is closed at filing. Open-ended "examine what is interesting in the data" exploration is explicitly not a V3 deliverable.

8. Infrastructure Pre-Flight Gate

Before the registered run, a mandatory end-to-end dry run is executed on $n=10$ dummy items, exercising: all five rounds, all five confirmatory-pool models, the JSON validation layer, and the scoring pipeline. The dry run also verifies that Condition A and Condition B execute as independent stateless runs with no leaked context, and verifies API keys, rate limits, and quota allocations for all models and the model-scorer service.

Any pipeline exception, schema-format failure, API timeout, or quota problem during the dry run triggers an immediate halt, a fix, and a restart of the dry run — NOT an amendment to the study.

Malformed-output rule. A response failing schema validation receives exactly one retry. (Deterministic sampling at temperature 0.0 makes further retries low-value; the retry primarily catches transient API faults.) If the response still fails, that model's response for that item is recorded as missing data. All malformed responses and retries are logged, and the malformed rate is a reported figure.

Halt rule: if malformed responses exceed 5% on a per-model basis, the run halts as a pipeline failure rather than producing a degraded dataset.

9. Order of Operations (Gating Sequence)

The pre-registration is filed only after every pre-filing step below is complete and no item remains open.

Completed at time of this draft: - Contrast-matched effect size computed from V2 data: $d = 0.64$ (Section 6.2). - Power analysis run; sample size fixed at $N = 80$ (Section 6.3). - Round protocol prompts and JSON schemas specified (Section 3.3). - Condition A/B causal design specified (Section 3.2). - Statistical tests and FDR procedure specified (Section 6). - Exploratory-analysis registry finalized (Section 7). - Scoring architecture, reliability tiers, and CVS rubric supplement specified and drafted (Section 5; rubric supplement file).

Remaining before filing: 1. Confirm the effect-size floor value ($d \geq 0.55$) — currently provisional pending external methodological review (Sections 2.1, 6.1). 2. Complete the cold-pass verification that the CVS rubric matches V2's applied criteria (Section 5.1). 3. Confirm filing venue and record timestamp + content hash (Section 12).

At/after filing (not gating the prereg itself): 4. File the pre-registration once the remaining items above are closed. 5. Configure and pass the infrastructure pre-flight gate (Section 8). 6. Run the pilot, which sets the Delta- μ / Delta- σ equivalence bounds per the pre-registered procedure (Section 6.6). 7. Begin the registered data collection.

Note: The Delta- μ / Delta- σ equivalence bounds (Section 6.6) are set from pilot data per a pre-registered procedure; the procedure is fixed before filing, the bound values are set during the pilot. This is a pre-specified, disclosed sequence, not a post-hoc decision.

Note: The Design 5 reversal analysis is owed for the final paper's limitations/ appendix discussion but, because no QPU component is included in V3 (Section 10), it does not gate filing.

10. Scope Exclusions

The following are explicitly OUT of scope for V3 and are recorded as candidates for future studies:

- **QPU-seeded routing.** Deferred. V3 is entirely classical. A quantum arm is deferred to a future study, pending full digestion of the Design 5 hardware-reversal confound stack.
- **Multi-model coordination performance.** Whether a council of models collectively reasons better than a single model is the broader IRMB program question. V3 measures individual-model behavior under a deliberation protocol; it does not test combined-system performance.
- **Context-length degradation of reasoning.** A candidate future confirmatory study; not tested by V3's design.
- **"Structural" fourth revision-trigger category.** Deferred to preserve V2 comparability (Section 3.3).

11. Pre-Registration Deviations Policy

V3 commits to a single pre-registration with zero post-collection amendments. Every foreseeable contingency is handled by a pre-registered branch (tiered human validation, reliability-failure branch, malformed-output rule, pre-flight failure). If an unforeseen issue arises that is not covered by a pre-registered branch, it will be reported transparently as a deviation with full disclosure — but the design intent is that no such deviation should be necessary.

12. Integrity

The pre-registration will be filed on the **Open Science Framework (OSF)** as an open-ended registration, which timestamps the document on OSF's servers and issues a permanent DOI. OSF is chosen over a self-hosted hash because independent third-party timestamping is what substantiates the confirmatory claim — it establishes that the pre-registration preceded data collection in a way a self-hosted record cannot. [TODO – at filing: record the OSF registration DOI, timestamp, and content hash here.]

Outstanding TODO Summary (must all be closed before filing)

- Section 6.6: predeclare Delta-mu and Delta-sigma equivalence bounds with rationale (set with Billy)
- Cold-pass verification: confirm rubric matches V2 applied CVS criteria (Section 5.1)
- At filing: record OSF DOI, timestamp, and content hash (Section 12)

Full Force Eternal | Romans 8:28